

JACK BELLEVILLE

Full-Stack Software Engineer | AI-Integrated Applications

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EDUCATION

University of Missouri - Columbia

BS in Computer Science, Minor in Business, Honors Scholar – 3.93/4.0 GPA

Columbia, MO

Expected Graduation May 2028

EXPERIENCE

Tonetta.ai

Dec. 2025 – Present

Founding Engineer

Columbia, MO

- Drove development of an automatic audio modification engine from concept to MVP, delivering <10ms real-time speech enhancement and achieving up to 15% improvement in user-perceived confidence scores.
- Architected end-to-end identity and subscription infrastructure using Supabase and Google OAuth 2.0, prioritizing secure access control and delivered through a responsive web dashboard.
- Contributed to product strategy, team operations, and fundraising efforts, helping close a \$27,500 pre-seed round at a \$1M valuation.
- Designed and implemented a scalable end-to-end onboarding funnel, driving a 10x increase in demo bookings and outreach conversions.

MAVI Computer Vision Lab, University of Missouri

Aug. 2025 – Present

Research Fellowship

Columbia, MO

- Developed computer vision methods for automated classification and diagnosis of pediatric bone underdevelopment and osteoarthritis from radiographic imaging.
- Annotated and curated training/validation datasets; engineered pre-processing pipelines to diagnose and resolve data quality issues, improving model accuracy by 20%.
- Conducted systematic benchmarking of PointNet++ on 3D soybean point cloud data, establishing performance baselines that validated a 200%+ efficiency gain in the resulting novel architecture.

AfterQuery Experts

Mar. 2026 – Present

AI Training Data Specialist

Remote

- Curated and annotated high-quality training datasets for AfterQuery, supporting AI model development at leading AI labs through rigorous data quality standards and domain expertise.

PROJECTS

Tonetta Backend Development | FastAPI, OAuth 2.0, Middleware

- Designed and implemented RESTful API endpoints in FastAPI, enforcing secure authentication middleware and input validation to protect platform integrity.
- Built bidirectional WebSocket communication pipelines to enable real-time audio processing, ensuring reliable performance under concurrent user load.

Pokémon Card Scanner | Python, OpenCV, SQLite, REST APIs

- Built a real-time card identification system using OpenCV perspective correction and four perceptual hashing algorithms, achieving orientation-invariant matching against a SQLite database across multiple 100+ card datasets.
- Integrated the Scrydex TCG API to display live market pricing for scanned cards, caching responses in SQLite with daily expiry to conserve metered API credits.

Space Ball | C#, Unity, VR

- Developed an immersive virtual reality game alongside a 3-person team where players guide a ball through rhythmic obstacle courses.
- Used C# to implement core gameplay systems and debug mechanics, and Unity to build an interactive 3D environment that is optimized to ensure stable performance on hardware-constrained standalone VR headsets.

TECHNICAL SKILLS

Languages: C, C#, C++17, Java, Python, TypeScript, JavaScript, SQL

ML & AI: PyTorch, TensorFlow, CUDA, ONNX Runtime, OpenCV, Open3D, LangChain, HuggingFace, Pandas, NumPy

Infrastructure: Git, Docker, Kubernetes, GCP, AWS, Unity, TensorDock, FastAPI, REST API, Supabase, Linux

Concepts: Point Clouds & 3D Convolution, Distributed Systems Fundamentals, RESTful API Design, Agile Development